

# Statistics And Probability: Hypothesis Tests and Control Charts

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## Test Of Hypothesis

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A **Hypothesis Test** is a formal procedure for deciding between two competing ideas (hypotheses) about a **population** based on evidence collected from a **sample**.

Imagine a large factory produces light bulbs (**the population**). A worker claims the average lifespan of these bulbs is 1,000 hours (**the hypothesis**). To check this claim, you take a small batch of bulbs (**the sample**) and measure their average lifespan.

- **The Goal:** We want to know if the information (the claim) about the population characteristic (average lifespan) can be **accepted** or **rejected**.
- **The Method:** We collect data from a small, representative part (the **sample**) and use that data to make a confident conclusion about the entire **population**.



## Symbols Of Population and Sample

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In statistics, we use different symbols to distinguish between information about the large population and information gathered from the small sample.

| Population Parameter/Characteristic | Sample Statistic   | Description                                                         |
|-------------------------------------|--------------------|---------------------------------------------------------------------|
| Population Size( $N$ )              | Sample Size( $n$ ) | The total number of items in the population vs. the number of items |

| Population Parameter/Characteristic       | Sample Statistic                 | Description                                                                                            |
|-------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------|
|                                           |                                  | collected for testing.                                                                                 |
| Population Mean( $\mu$ )                  | Sample Mean( $\bar{x}$ )         | The true average of the whole population (usually unknown) vs. the average calculated from the sample. |
| Population Standard Deviation( $\sigma$ ) | Sample Standard Deviation( $S$ ) | The spread or variability of the whole population vs. the spread calculated from the sample.           |
| Population Variance( $\sigma^2$ )         | Sample Variance( $S^2$ )         | The square of the standard deviation.                                                                  |



## Procedure Of Test Of Hypothesis

A hypothesis test follows a set of structured steps to ensure the decision is logical and statistically sound.

1. **State Null Hypothesis ( $H_0$ ) & Alternate Hypothesis ( $H_1$ ):** Formulate the two competing claims.
2. **Define Level of Significance ( $\alpha$ ):** Decide how much risk of being wrong you are willing to accept.
3. **Use Suitable Test Statistic:** Choose the correct formula (e.g.,  $z$ -test,  $t$ -test) to crunch the sample numbers.
4. **Establish Critical Region (Table Value):** Determine the cut-off point for the test statistic, based on the chosen significance level.

5. **Conclusion:** Compare your calculated result to the cut-off point and make the decision to **Accept** or **Reject** the Null Hypothesis.

## Null Hypothesis ( $H_0$ )

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The **Null Hypothesis** ( $H_0$ ) is the initial assumption or the 'status quo' that we assume to be true. It always states that there is **no significant difference** or **no change** in the population parameter.

- **Assumption:** We start by assuming that the sample result ( $\bar{x}$ ) is just a fluke and the population parameter ( $\mu$ ) is exactly as claimed.
- **Wording:** It always contains an equality sign ( $=$ ,  $\leq$ , or  $\geq$ ).
  - **Example:**  $H_0$ : The average lifespan of a light bulb is 1,000 hours ( $\mu = 1000$ ).

## Alternative Hypothesis ( $H_1$ )

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The **Alternative Hypothesis** ( $H_1$ ) is the statement we hope to prove. It contradicts the null hypothesis and asserts that a **significant difference** or **change does** exist.

- **Wording:** It always contains an inequality sign ( $\neq$ ,  $>$ , or  $<$ ).
  - **Example (Claim of Change):**  $H_1$ : The average lifespan is **not** 1,000 hours ( $\mu \neq 1000$ ).

## Single-Tailed vs. Two-Tailed Test

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The form of the alternative hypothesis determines whether we use a one-tailed or two-tailed test. Let's assume  $H_0 : \mu = \mu_0$  (where  $\mu_0$  is the assumed value).

| Alternate Hypothesis ( $H_1$ ) | Type of Test                      | Focus of Rejection                                                                   |
|--------------------------------|-----------------------------------|--------------------------------------------------------------------------------------|
| $\mu \neq \mu_0$               | <b>Two-Tailed Test</b>            | Rejection can happen if $\mu$ is significantly <b>higher or lower</b> than $\mu_0$ . |
| $\mu > \mu_0$                  | <b>Single (Right) Tailed Test</b> | Rejection only happens if $\mu$ is significantly <b>higher</b> than $\mu_0$ .        |

| Alternate Hypothesis ( $H_1$ ) | Type of Test                     | Focus of Rejection                                                           |
|--------------------------------|----------------------------------|------------------------------------------------------------------------------|
| $\mu < \mu_0$                  | <b>Single (Left) Tailed Test</b> | Rejection only happens if $\mu$ is significantly <b>lower</b> than $\mu_0$ . |

## Level of Significance ( $\alpha$ )

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The **Level of Significance** ( $\alpha$ ) is the maximum probability of **rejecting the Null Hypothesis ( $H_0$ ) when it is actually true**. This is a risk we accept, and it is set *before* the test begins.

- $\alpha$  is often called the probability of a **Type I Error** (false positive).
- **Common Values:**  $\alpha = 0.05$  (or **5%**) and  $\alpha = 0.01$  (or **1%**).
  - If  $\alpha = 5\%$ , you are willing to be wrong 5 out of 100 times when you decide to reject  $H_0$ .

## Critical Value and Critical Region

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### Critical Region (or Rejection Region)

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The **Critical Region** is the area in the distribution (usually a Normal or  $t$ -distribution) where, if the **calculated test statistic** falls, we must **Reject the Null Hypothesis ( $H_0$ )**. This area corresponds to the  $\alpha$  level.

### Critical Value

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The **Critical Value** is the boundary line (a specific numerical value) that separates the Critical Region (Rejection Region) from the **Acceptance Region**.

The value depends on:

1. **Level of Significance ( $\alpha$ ):** How large the rejection area is.

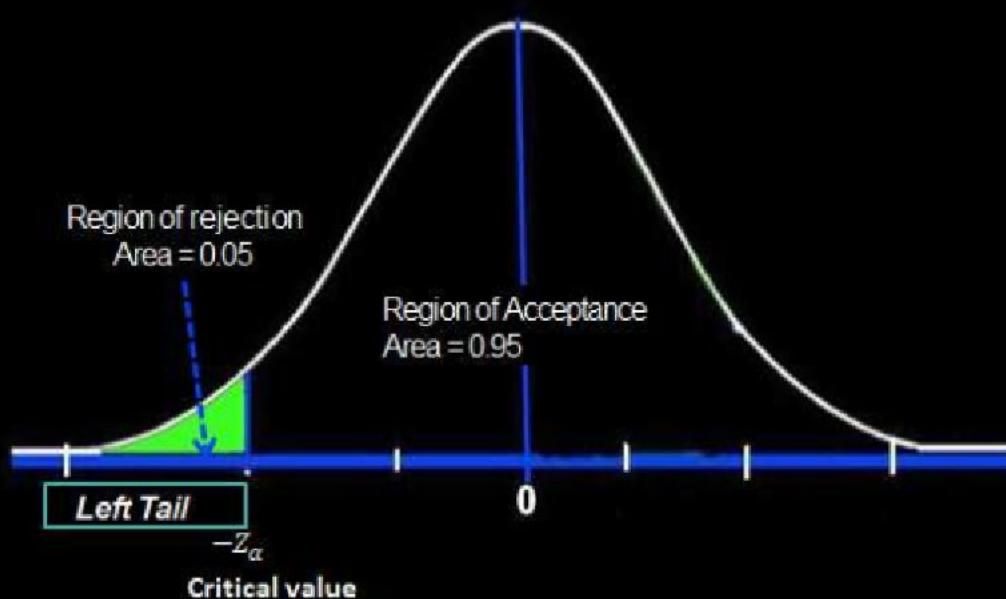


2. **Alternative Hypothesis ( $H_1$ )**: Whether the area is split into two tails or concentrated in one tail (left or right).

If  $\alpha = 5\%$  (0.05) and we use a Z-test (large sample):

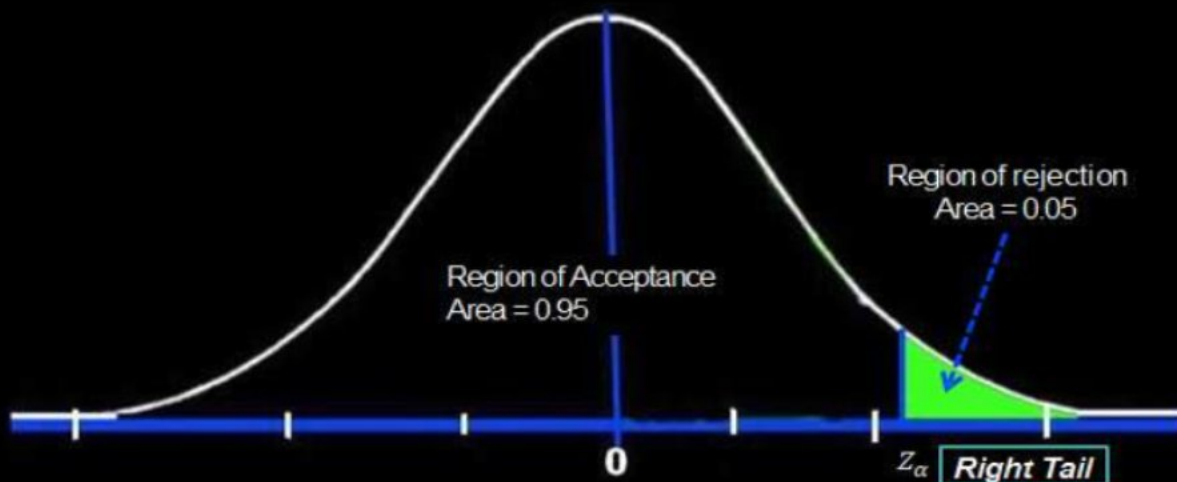
| Test Type           | $H_1$            | Critical Value(s) (Z-Scores)      |
|---------------------|------------------|-----------------------------------|
| Two-Tailed          | $\mu \neq \mu_0$ | $\pm 1.96$ (0.025 in each tail)   |
| Single Left-Tailed  | $\mu < \mu_0$    | $-1.645$ (0.05 in the left tail)  |
| Single Right-Tailed | $\mu > \mu_0$    | $+1.645$ (0.05 in the right tail) |

Critical region is the region in which a sample value is rejected. Single left tailed test  $\alpha = 5\%$



Single Right Tailed Test If  $\alpha = 5\%$

## Single right tailed test if $\alpha = 5\%$



Two Tailed Test If  $\alpha = 5\%$

## Two Tailed Test if $\alpha = 5\%$



## Conclusion

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The final step is to compare the value of the **Test Statistic (Calculated Value)** to the **Critical Value (Table Value)**.

- **Decision Rule:**

- **If**  $|\text{Calculated Value}| \leq |\text{Critical Value}|$ : The calculated value falls in the **Acceptance Region**. We **Accept**  $H_0$  (or **Fail to Reject**  $H_0$ ). The evidence is not strong enough to reject the initial assumption.
- **If**  $|\text{Calculated Value}| > |\text{Critical Value}|$ : The calculated value falls in the **Critical/Rejection Region**. We **Reject**  $H_0$  and **Accept**  $H_1$ . The evidence strongly suggests the initial assumption is false.